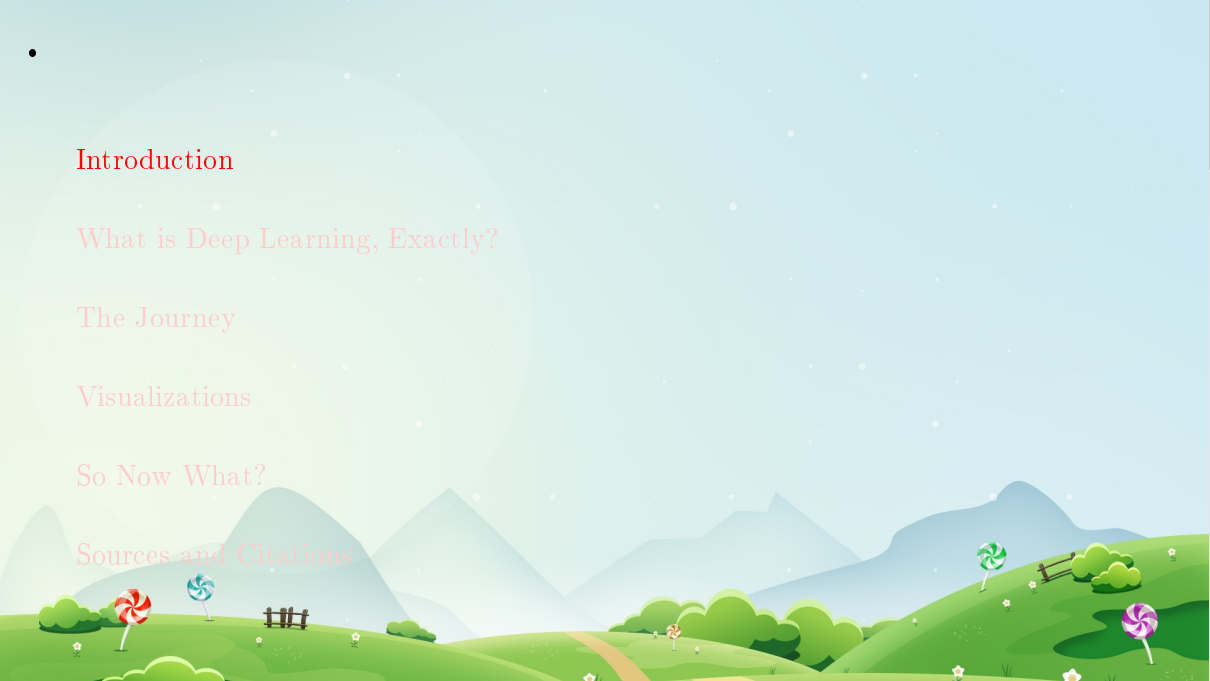


Learning About Deep Learning, and Maybe a Few Other Things

Franklin Diaz, Cosmic Voyager

2022-09-27

- My presentation title is here.
- This talk is about my on-going journey into the world of graph theory and neural networks.
- This is a description of the learning process as well as the project and presenting some results.



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└ Introduction

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This first section is an Introduction to me and my project

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About Me

- I am a Security Consultant at Palo Alto Networks, cloud and automation for past 2+ years
- Did Data Eng/DevSecOps at Salesforce for 5 years.
- Been going to security conferences for a while.



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Introduction

- Here is a picture of me, modified by a popular local artist.
- In my current role as a consultant, I get to work with the major cloud providers.
- In the past I was not a Data Scientist, but did some time on the Security Data Engineering team at Salesforce. This gave me a bit of a head start with data pipelines, directed acyclic graphs, and a few other things.

- I am a Security Consultant at Palo Alto Networks, cloud and automation for past 2+ years
- Did Data Eng/DevSecOps at Salesforce for 5 years.
- Been going to security conferences for a while.



The Project

- Realized that Terraform can output directed graphs.
- Had done a lot of work at Salesforce with directed graphs, data pipeline orchestration with AirFlow, etc. so I was somewhat familiar with the output I was seeing.
- The first question I had was, what can I do with these directed graphs?
- My hunch was I could “do some processing and analysis” of all this security infrastructure graph data and hoped that could lead to... predictions?
- [All of the Code for \(almost\) everything is Here](#)

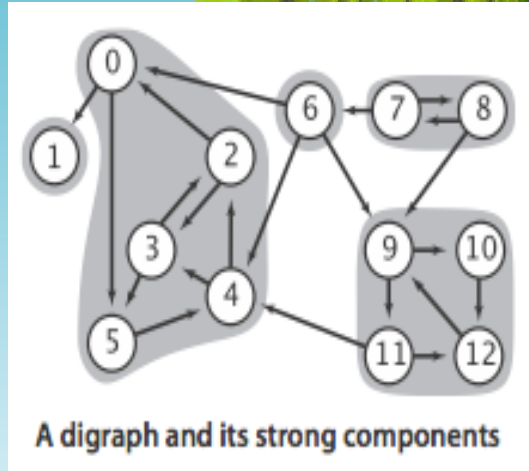
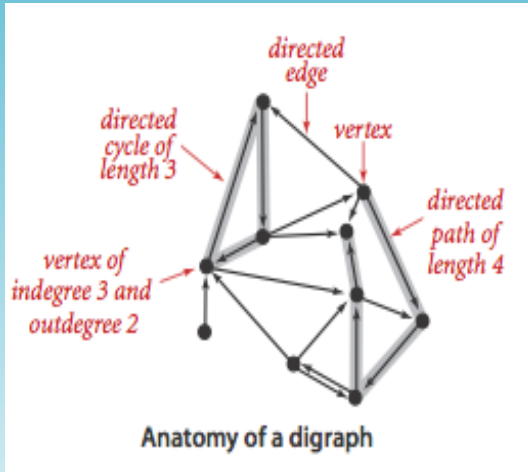
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- In case you are not familiar, Terraform is software that allows you to declare resources like network elements in public cloud providers.
- Had and still have this vague notion that if I had enough data I could find “outliers”. Maybe like a modernized version of a Pareto analysis?

- Realized that Terraform can output directed graphs.
- Had done a lot of work at Salesforce with directed graphs, data pipeline orchestration with AirFlow, etc. so I was somewhat familiar with the output I was seeing.
- The first question I had was, what can I do with these directed graphs?
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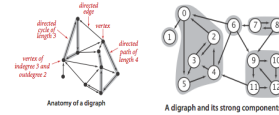
What's a DiGraph?



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Introduction

- The big takeaway here is the idea of “edges” and “nodes”
- Source: Algorithms, 4th Edition, by Robert Sedgewick and Kevin Wayne
- Wrath of Math!





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└─ What is Deep Learning, Exactly?

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- You’ve probably heard this term lately, lets talk about what it means.

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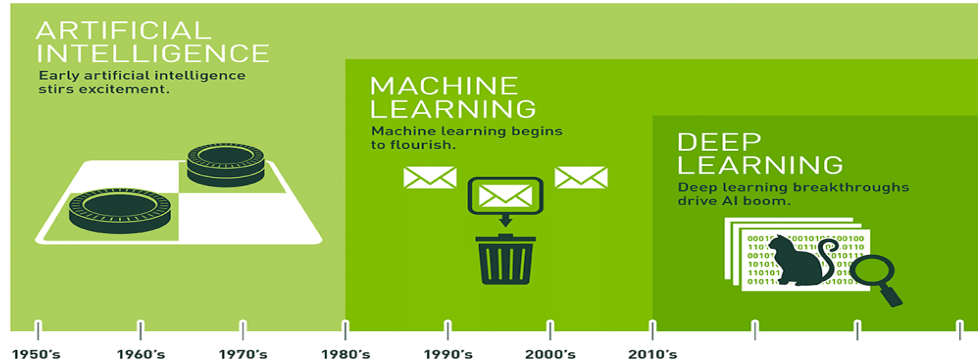
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The Rise of Deep Learning



Since an early flush of optimism in the 1950s, smaller subsets of artificial intelligence – first machine learning, then deep learning, a subset of machine learning – have created ever larger disruptions.

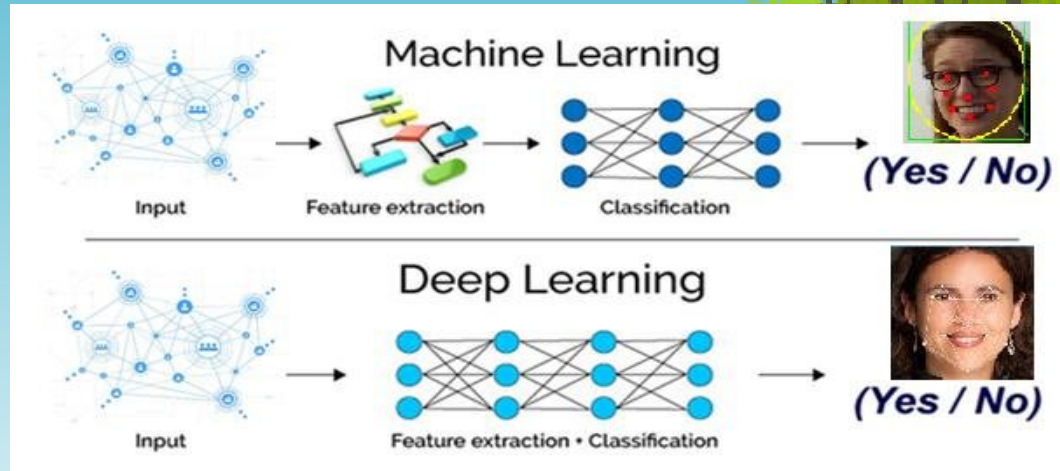
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What is Deep Learning, Exactly?

- GPUs have made it possible to expand accessibility to DL
- the CUDA toolkit from Nvidia has made things easier for researchers.

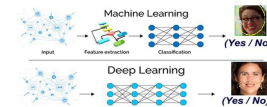


Quick Intro to a Giant Topic



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What is Deep Learning, Exactly?



- [image source/credit](#)
- ML feature extraction can be a huge undertaking, up to 80% of a project.
- DL attempts to automatically learn features that are most useful for a task from raw data.
- The nodes in a digraph are “neurons” or “units” in the DL/graph theory context.
- The neurons perform two steps. They calculate a “weighted sum” and pass the result through an “activation function” such as a rectifier activation function.
- These neurons or units that go through the rectifier function are called “RelUs” for short. Lot’s of descriptive info in this one term!
- Depth of the GNN is measured by the number of connected layers.
- DL needs very large data sets for accurate feature determination. Data sets with lots of features are known as “high density”.
- We humans interpret the features and output based on what we are trying to model.

Amazing Training and Tools Available

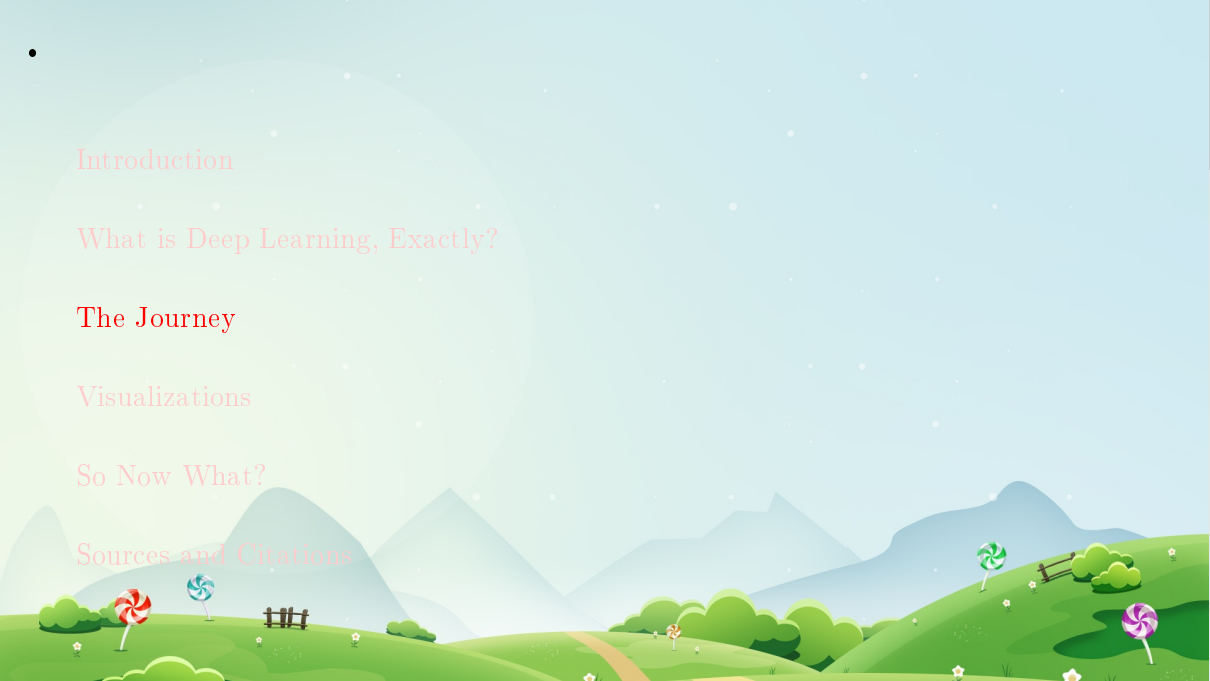
- There is a ton of information suddenly. Books, papers, code, etc.
- Folks are very helpful, positioning themselves as experts.
- [super helpful videos like this one](#)
- The [Google Machine Learning Crash Course](#) is free with tons of information.

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└─ What is Deep Learning, Exactly?

- Google Deep Learning Container Images
- Continuous Machine Learning (CML) Project
- Kaggle and shared Jupyter Notebooks

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- Now I would like to talk a bit about the shape of the project.

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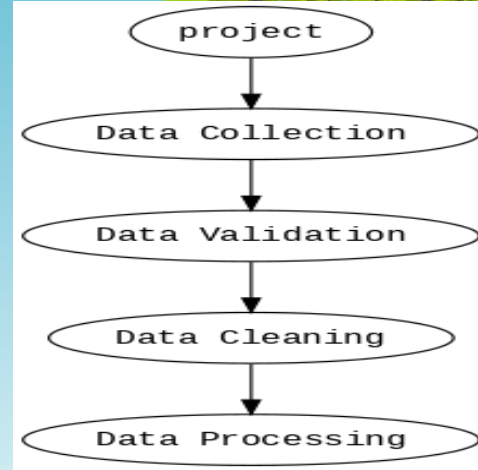
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What Have I Gotten Myself Into?

- It didn't take long before I realized the magnitude of the ocean I was wading into.
- Started reading everything I could find even though I didn't understand most of it.
- I came up with the basic framework you see in the image here.



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The Journey

- Repetition can be a slow and painful way to learn.
- Wasn't even sure what questions to ask. Slow going at first.

- It didn't take long before I realized the magnitude of the ocean I was wading into.
- Started reading everything I could find even though I didn't understand most of it.
- I came up with the basic framework you see in the image here .



Yak Shaving, Side Quests, Endless Rabbit Holes

- Makefiles and GNU Autotools
- NVIDIA Jetson Nano as cluster nodes
- SLURM cluster scheduler
- OpenMPI for parallel builds
- Docker and Containers
- k8s and Rancher k3s
- Data Version Control dvc.org
- Storing/accessing data in GCP buckets
- Continuous Machine Learning cml.dev
- Internal PyPI and Debian/Raspbian mirror (used too much bandwidth on home connection)

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The Journey

- Wasn't sure exactly where to drop this slide in the order.
- Trying to show that there have been a TON of side quests.
- Some of these were useful, some led to spin off projects. A lot of this is bookmarked for later when I get some “spare time” haha.

- Makefiles and GNU Autotools
- NVIDIA Jetson Nano as cluster nodes
- SLURM cluster scheduler
- OpenMPI for parallel builds
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- Internal PyPI and Debian/Raspbian mirror (used too much bandwidth on home connection)

Dot Data Collection

- A big barrier to entry was removed by the ability to output a Directed Graph from Terraform.
- [Click for video](#)

```
# Generate a PNG from Terraform  
terraform graph | dot -Tpng > graph.png
```

```
# Generate vector graphic from Terraform  
terraform graph | dot -Tsvg -o graph.svg
```

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The Journey

- Was pretty happy I could generate a PNG file. Super easy!
- Then I opened up the file and took a look at the nodes in the graph....

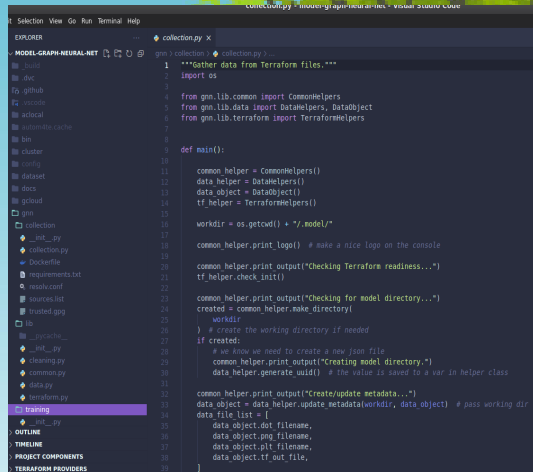
• A big barrier to entry was removed by the ability to output a Directed Graph from Terraform.

• [Click for video](#)

```
# generate a png from terraform  
terraform graph | dot -Tpng > graph.png  
  
# generate vector graphic from terraform  
terraform graph | dot -Tsvg -o graph.svg
```

Python Data Collection

- This became the basis for data collection via Python.
- Found a cool module on Pypi called **python-terraform** that allowed me to run Terraform CLI commands from inside Python.
- [Click for video](#)



```
1 """Gather data from Terraform files."""
2 import os
3
4 from gnn.lib.common import CommonHelpers
5 from gnn.lib.data import DataHelpers, DataObject
6 from gnn.lib.terraform import TerraformHelpers
7
8
9 def main():
10
11     common_helper = CommonHelpers()
12     data_helper = DataHelpers()
13     data_object = DataObject()
14     tf_helper = TerraformHelpers()
15
16     workdir = os.getcwd() + "/.model/"
17
18     common_helper.print_logo() # make a nice logo on the console
19
20     common_helper.print_output("Checking Terraform readiness...")
21     tf_helper.check_init()
22
23     common_helper.print_output("Checking for model directory...")
24     created = common_helper.make_directory(
25         workdir
26     ) # create the working directory if needed
27     if created:
28         # we know we need to create a new json file
29         common_helper.print_output("Creating model directory.")
30         data_helper.generate_uuid() # the value is saved to a var in helper class
31
32     common_helper.print_output("Create/update metadata...")
33     data_object = data_helper.update_metadata(workdir, data_object) # pass working dir
34     data_file_list = [
35         data_object.dot_filename,
36         data_object.png_filename,
37         data_object.plt_filename,
38         data_object.tf_out_file,
39     ]
```

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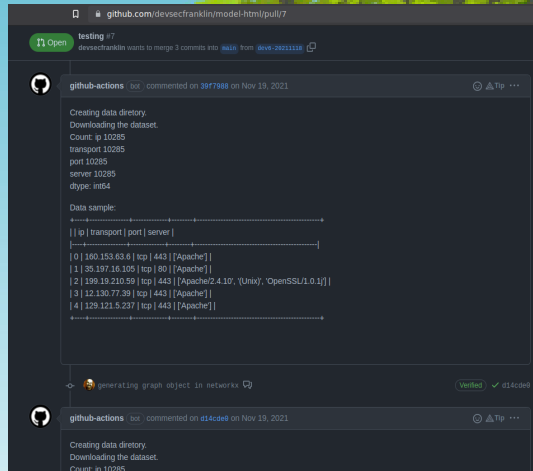
- Kind of a no brainer.
- The video is sped up 3x or so, but you can get the flavor of how the project looks from this.

- This became the basis for data collection via Python.
- Found a cool module on Pypi called **python-terraform** that allowed me to run Terraform CLI commands from inside Python.
- [Click for video](#)



Data Processing Side Quest

- This is what happens when you spend a week with P0lr.



The screenshot shows a GitHub pull request for 'model-html' with the title 'testing #7'. The pull request is from 'dev-2621138' to 'main'. The workflow run 'github-actions' is shown with the following steps:

```
Creating data directory.
Downloading the dataset.
Count: ip 10285
transport 10285
port 10285
server 10285
dtype: int64

Data sample:
+-----+
| ip | transport | port | server |
+-----+
| 0 | 160.153.63.6 | tcp | 443 | [Apache] |
| 1 | 35.197.16.105 | tcp | 80 | [Apache] |
| 2 | 199.19.210.59 | tcp | 443 | [Apache/2.4.10', 'Unix', 'OpenSSL/1.0.1]] |
| 3 | 12.130.77.39 | tcp | 443 | [Apache] |
| 4 | 129.121.5.237 | tcp | 443 | [Apache] |
+-----+
```

The workflow run is verified and completed successfully.

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The Journey

- Spent a week with P0lr where we had a moderate case of machine learning fever.
- We had a direction but no destination.
- Watched Alpha Go movie, talked about a bunch of stuff, read some books and papers.
- wound up writing some code.
- there was a “HTML model” in there somewhere too

- This is what happens when you spend a week with P0lr.



Data Storage - Kaggle

- What the heck is it?



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- What the heck is it?

• What the heck is it?

Data Storage - Google Cloud

Data storage with GCP because it's (relatively) easy.



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Data storage with GCP because it's (relatively) easy.

Data Storage - DVC

- Data storage and tagging using DVC
- there is a video on this page that explains



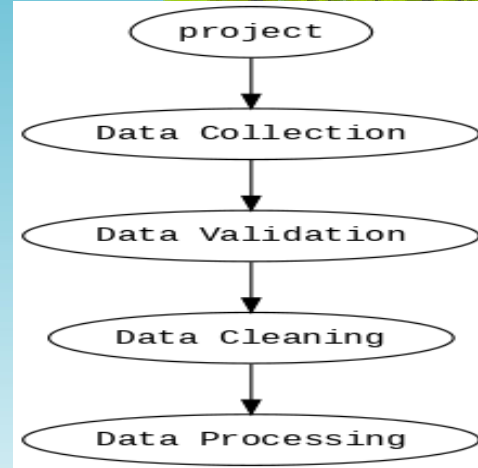
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- Data storage and tagging using DVC
- there is a video on this page that explains

Data Pipeline

- The Data Pipeline is a set of processes that move and transform data from various sources to a destination where new value can be derived.
- The DP is the foundation of analytics, reporting, and machine learning capabilities.



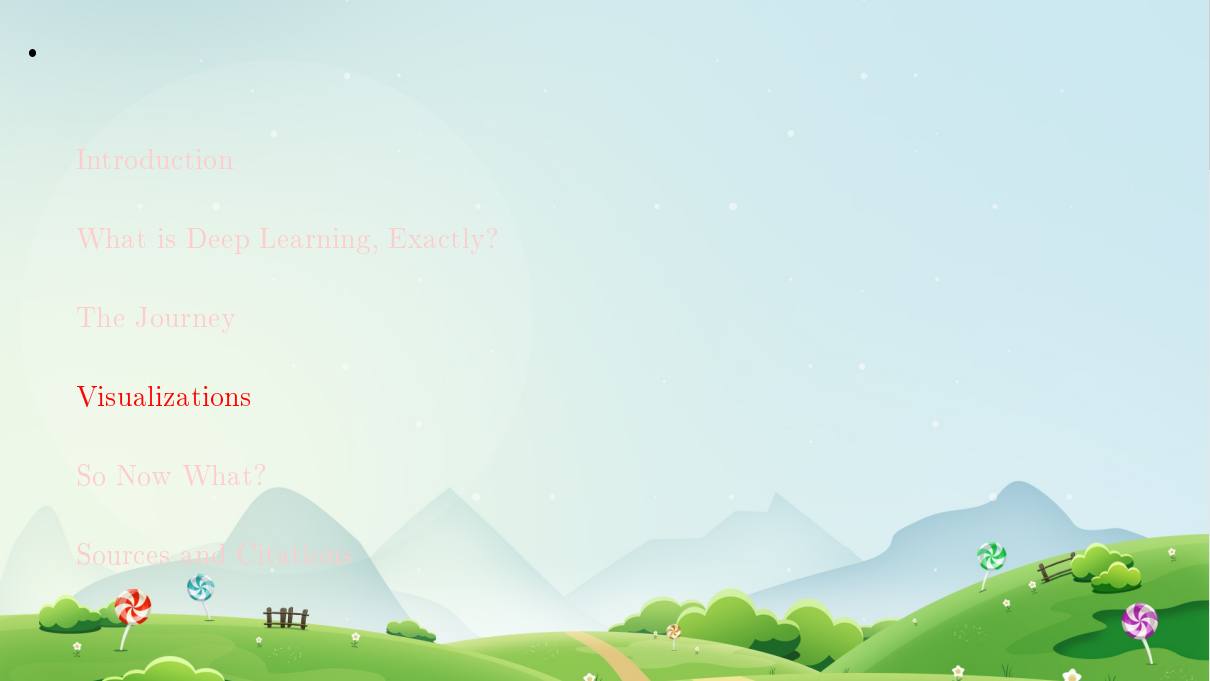
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- Source: Data Pipelines pocket reference p1-2

- The Data Pipeline is a set of processes that move and transform data from various sources to a destination where new value can be derived.
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[illegible]

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└ Visualizations

- This is the first thing I saw when I started converting the data.
- Was excited here since I was able to change the color of the nodes.
- Obviously this is not yet a usable result
- **some video of data collection**

- Let G be a graph with n vertices that are assumed to be ordered from

The Adjacency Matrix

to

v_1

v_n

The $n \times n$ matrix A , in which

$$a_{ij} = 1$$

if there exists a path from

v_i

to

v_j

$$a_{ij} = 0$$

otherwise
is called an adjacency matrix.

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└ Visualizations

• Let G be a graph with n vertices that are assumed to be ordered from

v_1
 v_2
 $\vdots$
 v_n

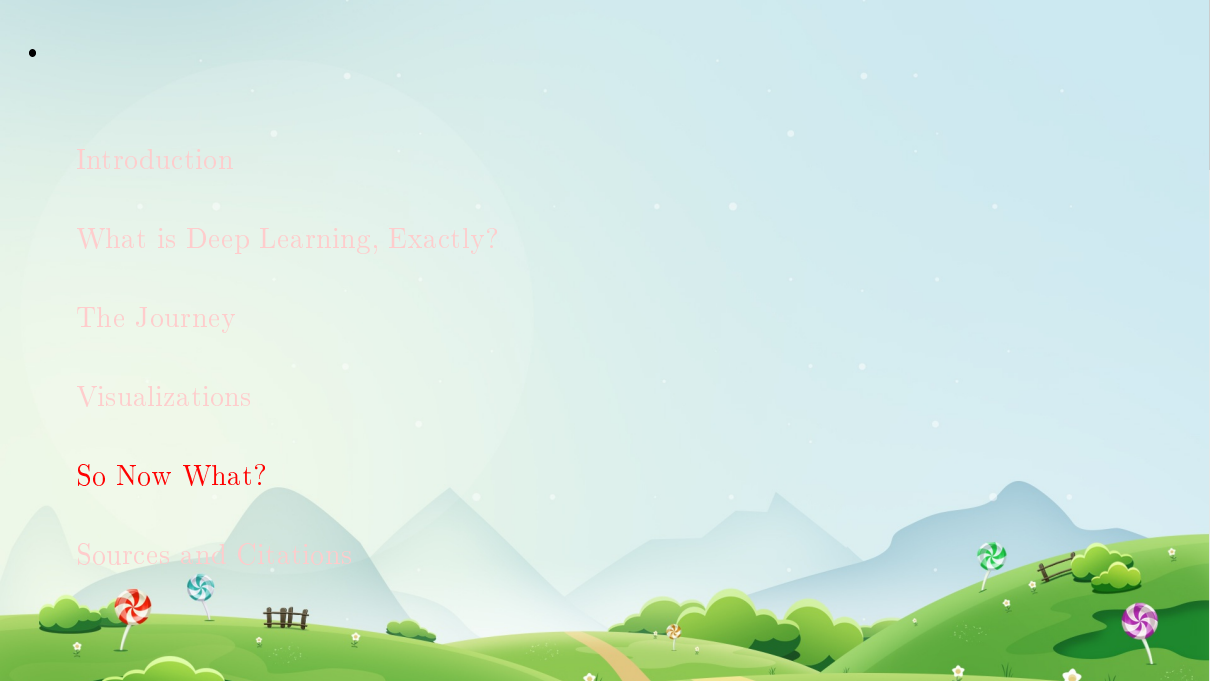
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 v_i
 v_j
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Useful Intermediate Results

- Standardizing my data collection on JSON.
- Made some super cool functions for parsing nested JSON.
- Turned some of this time into money with **cloud tools**
- Importing JSON to Pandas dataframes.

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└ So Now What?

- Tabular data in Pandas can be output in all kinds of formats.
- Pandas data frames can be the input for other Machine Learning tools and frameworks.

- Standardizing my data collection on JSON .
- Made some super cool functions for parsing nested JSON .
- Turned some of this time into money with **cloud tools**
- Importing JSON to Pandas dataframes .

Next Steps for this Project

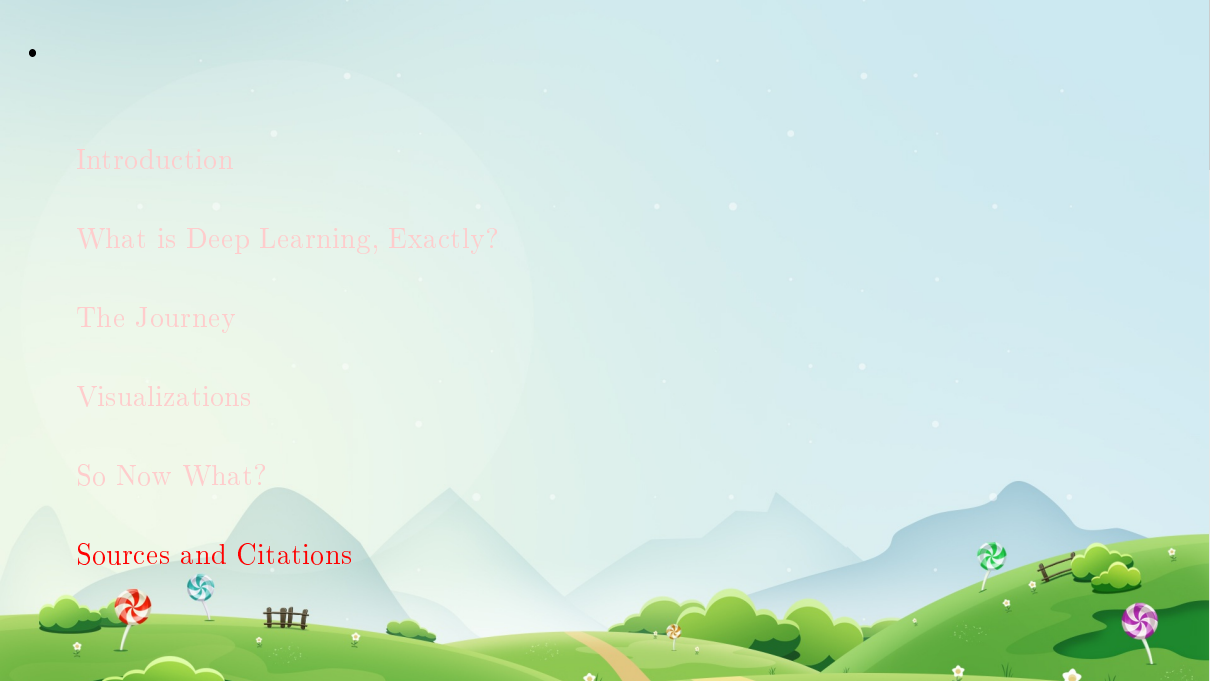
- The data collection problem (Don't have access to enough data!)
- Maybe have a “collection container” with a python/Flask RESTful API for folks to push data to? Or even better, scrape GH for public repos with Terraform? (Lots of data, but not all security infra)
- Maybe back to Kaggle to find some big data to operate on?
- See if I can get the training to use my personal GPU/TPU.

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So Now What?

- Most of this work is relegated to my “free” time.
- Have to spend my days helping people with the cloud.

- The data collection problem: Don't have access to enough data!
- Maybe have a “collection container” with a python/Flask RESTful API for folks to push data to? Or even better, scrape GH for public repos with Terraform? (Lots of data, but not all security infra)
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Sources and Citations

- Generate a bibtex

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- Generate a bibtex

- Need to add a reading list on here
- Make a list of all the books and papers